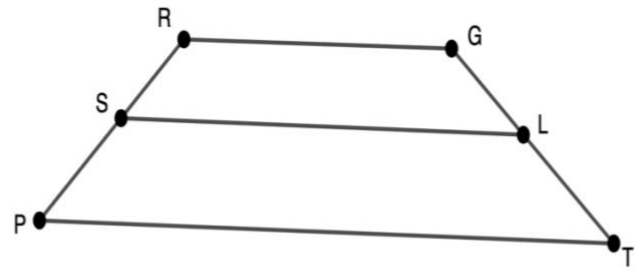
Geometry
Unit 8
Lesson 7 Practice

Name____________________
Date____________________
Period____________________

$PTGR$ is a trapezoid as shown where $\overline{PT} \parallel \overline{RG}$. $\overline{SL}$ is the midsegment of $PTGR$ with $SL = 18$, $PT = 9x - 5$, and $RG = 7x - 3$.

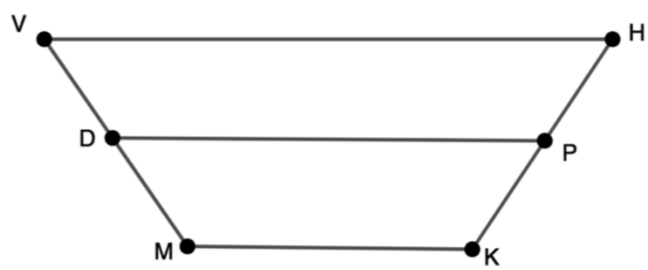


1. Write an equation to show the relationship between $\overline{SL}$ and the bases of the trapezoid.

2. What is the value of x? Round your answer to the nearest hundredth.

3. What are the lengths of $\overline{PT}$ and $\overline{RG}$? Round your answers to the nearest hundredth.

Trapezoid $HKMV$ is shown with midsegment $\overline{DP}$ where $VH = 9x + 12,\ MK = 6x,$ and $DP = 10x$.

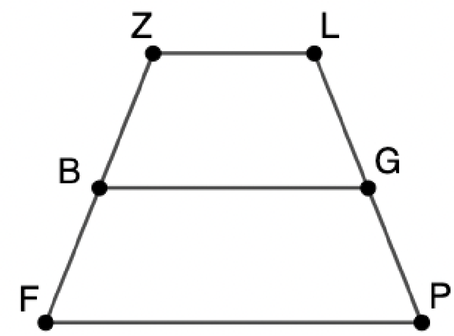


4. What is the value of x? Round your answer to the nearest tenth.

5. What is the length of $\overline{DP}$?

6. What are the lengths of $\overline{VH}$ and $\overline{MK}$? Round your answer to the nearest tenth.

Isosceles trapezoid $FPLZ$ is shown with midsegment $\overline{BG}$ where $ZL = 5x + 1,\ FP = 8x + 2,\ BG = 4x + 4,\ BZ = 3.5$.



7. What is the value of x?

8. What is the length of $\overline{BG}$?

9. What are the lengths of $\overline{ZL}$ and $\overline{FP}$?

10. What is the perimeter of $FPLZ$?